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Method for forming chip package structure

Abstract

A chip package structure is provided. The chip package structure includes a wiring substrate. The chip package structure includes an interposer substrate over the wiring substrate. The interposer substrate includes a redistribution structure, a dielectric layer, a conductive via, and a plurality of first dummy vias, the dielectric layer is over the redistribution structure, the conductive via and the first dummy vias pass through the dielectric layer, the first dummy vias surround the conductive via, and the first dummy vias are electrically insulated from the wiring substrate. The chip package structure includes a chip structure over the interposer substrate. The chip structure is electrically connected to the conductive via, and the chip structure is electrically insulated from the first dummy vias.

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Background/Summary

BACKGROUND

(1) Semiconductor devices are used in a variety of electronic applications, such as personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements.

(2) Dozens or hundreds of integrated circuits are typically manufactured on a single semiconductor wafer. The individual dies are singulated by sawing the integrated circuits along scribe lines. The individual dies are then packaged separately. The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area. However, since feature sizes of dies continue to decrease, feature sizes of an interposer substrate for carrying the dies decrease as well. Therefore, it is a challenge to form reliable packages with the dies and the interposer substrate with small feature sizes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1A-1H are cross-sectional views of various stages of a process for forming a chip package structure, in accordance with some embodiments.

(3) FIG. 1A-1 is a top view of a region of the redistribution structure of FIG. 1A, in accordance with some embodiments.

(4) FIG. 1C-1 is a top view of a region of the redistribution structure of FIG. 1C, in accordance with some embodiments.

(5) FIG. 1E-1 is a top view of a right region of the redistribution structure of FIG. 1E, in accordance with some embodiments.

(6) FIG. 1E-2 is a top view of a left region of the redistribution structure of FIG. 1E, in accordance with some embodiments.

(7) FIG. 1H-1 is a top view of the chip package structure of FIG. 1H, in accordance with some embodiments.

(8) FIG. 1H-2 is a top view of the chip package structure of FIG. 1H, in accordance with some embodiments.

(9) FIG. 1H-3 is a top view of the chip package structure of FIG. 1H, in accordance with some embodiments.

(10) FIG. 2 is a top view of a corner region of the interposer substrate of FIG. 1H, in accordance with some embodiments.

(11) FIG. 3 is a top view of a corner region of the interposer substrate of FIG. 1H, in accordance with some embodiments.

(12) FIG. 4 is a top view of a corner region of the interposer substrate of FIG. 1H, in accordance with some embodiments.

(13) FIG. 5 is a top view of a corner region of the interposer substrate of FIG. 1H, in accordance with some embodiments.

(14) FIG. 6 is a top view of a corner region of the interposer substrate of FIG. 1H, in accordance with some embodiments.

(15) FIG. 7 is a top view of a corner region of the interposer substrate of FIG. 1H, in accordance with some embodiments.

(16) FIG. 8 is a top view of a corner region of the interposer substrate of FIG. 1H, in accordance with some embodiments.

(17) FIG. 9 is a cross-sectional view illustrating a chip package structure, in accordance with some embodiments.

(18) FIGS. 10A-10B are cross-sectional views of various stages of a process for forming a chip

package structure, in accordance with some embodiments.

(19) FIG. **10A-1** is a top view of a corner region of the interposer substrate of FIG. **10A**, in accordance with some embodiments.

DETAILED DESCRIPTION

(20) The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(21) Furthermore, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(22) The term “substantially” in the description, such as in “substantially flat” or in “substantially coplanar”, etc., will be understood by the person skilled in the art. In some embodiments the adjective substantially may be removed. Where applicable, the term “substantially” may also include embodiments with “entirely”, “completely”, “all”, etc. The term “substantially” may be varied in different technologies and be in the deviation range understood by the skilled in the art. For example, the term “substantially” may also relate to 90% or higher, such as 95% or higher, especially 99% or higher, including 100%, though the present invention is not limited thereto. Furthermore, terms such as “substantially parallel” or “substantially perpendicular” may be interpreted as not to exclude insignificant deviation from the specified arrangement and may include for example deviations of up to 10°. The word “substantially” does not exclude “completely” e.g. a composition which is “substantially free” from Y may be completely free from Y.

(23) The term “about” may be varied in different technologies and be in the deviation range understood by the skilled in the art. The term “about” in conjunction with a specific distance or size is to be interpreted so as not to exclude insignificant deviation from the specified distance or size. For example, the term “about” may include deviations of up to 10%, though the present invention is not limited thereto. The term “about” in relation to a numerical value x may mean $x \pm 5$ or 10%, though the present invention is not limited thereto.

(24) Some embodiments of the disclosure are described. Additional operations can be provided before, during, and/or after the stages described in these embodiments. Some of the stages that are described can be replaced or eliminated for different embodiments. Additional features can be added to the semiconductor device structure. Some of the features described below can be replaced or eliminated for different embodiments. Although some embodiments are discussed with operations performed in a particular order, these operations may be performed in another logical order.

(25) FIGS. **1A-1H** are cross-sectional views of various stages of a process for forming a chip package structure, in accordance with some embodiments. FIG. **1A-1** is a top view of a region A of the redistribution structure **110** of FIG. **1A**, in accordance with some embodiments. The region A of

FIG. 1A shows a cross-sectional view illustrating the redistribution structure **110** along a sectional line A-A in FIG. 1A-1, in accordance with some embodiments.

(26) As shown in FIGS. 1A and 1A-1, a redistribution structure **110** is provided, in accordance with some embodiments. The redistribution structure **110** is used to from an interposer substrate for integrated fan-out packaging or wafer level packaging, in accordance with some embodiments. The redistribution structure **110** includes a dielectric layer **111**, redistribution layers **112**, **113**, **114** and **115**, and conductive vias **116**, in accordance with some embodiments. The redistribution layers **112**, **113**, **114** and **115** and the conductive vias **116** are formed in the dielectric layer **111**, in accordance with some embodiments.

(27) As shown in FIG. 1A, the conductive vias **116** are electrically connected between the redistribution layers **114** and **115**, in accordance with some embodiments. In some embodiments, conductive vias (not show) are electrically connected between the redistribution layers **112**, **113**, and **114**. For the sake of simplicity, FIG. 1A only shows four redistribution layers (i.e., **112**, **113**, **114** and **115**), in accordance with some embodiments. It should be noted that the number of redistribution layers is not limited in the present embodiment. In other words, there may be five or more redistribution layers.

(28) As shown in FIGS. 1A and 1A-1, the redistribution layer **114** includes conductive lines **114a**, in accordance with some embodiments. The redistribution layer **115** includes pads **115a** and **115b**, dummy pads **115c**, and conductive lines (not shown), in accordance with some embodiments. The conductive lines are connected to the pads **115b** (and **115a**), in accordance with some embodiments.

(29) The dummy pads **115c** surround the corresponding pad **115a**, in accordance with some embodiments. The width **W115c** of the dummy pad **115c** is substantially equal to or greater than the width **W115a** of the pad **115a**, in accordance with some embodiments. In some other embodiments, width **W115c** is less than the width **W115a**.

(30) In some embodiments, a dummy pad **115c** is between the pads **115a** and **115b**, and a distance **D1** between the pad **115a** and the dummy pad **115c** is less than a distance **D2** between the dummy pad **115c** and the pad **115b**. In some other embodiments, the distance **D1** is substantially equal to or greater than the distance **D2**.

(31) The distance **D1** ranges from about 25 μm to about 75 μm , in accordance with some embodiments. The distance **D2** ranges from about 25 μm to about 75 μm , in accordance with some embodiments.

(32) The conductive vias **116** are connected between the pads **115a** and the conductive lines **114a**, in accordance with some embodiments. In some embodiments, portions of the conductive lines **114a** are under the dummy pads **115c**. That is, the dummy pads **115c** cover or overlap the portions of the conductive lines **114a**, in accordance with some embodiments.

(33) The dielectric layer **111** is made of an insulating material such as a polymer material (e.g., polybenzoxazole, polyimide, or a photosensitive material), nitride (e.g., silicon nitride), oxide (e.g., silicon oxide), silicon oxynitride, or the like, in accordance with some embodiments.

(34) The dielectric layer **111** is formed using deposition processes (e.g. chemical vapor deposition processes or physical vapor deposition processes), photolithography processes, and etching processes, in accordance with some embodiments. The redistribution layers **112**, **113**, **114** and **115** and the conductive vias **116** are made of a conductive material, such as metal (e.g. copper, aluminum, or tungsten) or alloys thereof, in accordance with some embodiments.

(35) As shown in FIG. 1B, a dielectric layer **120** is formed over the redistribution structure **110**, in accordance with some embodiments. The dielectric layer **120** has openings **122**, **124**, and **126**, in accordance with some embodiments. The openings **122** expose the pads **115a** thereunder, in accordance with some embodiments. The openings **124** expose the pads **115b** thereunder, in accordance with some embodiments. The openings **126** expose the pads **115c** thereunder, in accordance with some embodiments.

(36) The dielectric layer **120** is made of an insulating material such as a polymer material (e.g.,

polybenzoxazole, polyimide, or a photosensitive material), nitride (e.g., silicon nitride), oxide (e.g., silicon oxide), silicon oxynitride, or the like, in accordance with some embodiments. The dielectric layer **120** is formed using deposition processes (e.g. chemical vapor deposition processes or physical vapor deposition processes), photolithography processes, and etching processes, in accordance with some embodiments.

(37) As shown in FIG. **1B**, a seed layer **132** is conformally formed over the dielectric layer **120** and the redistribution layer **115**, in accordance with some embodiments. The seed layer **132** is made of a conductive material, such as metal (e.g., copper, aluminum, gold, silver, or tungsten) or alloys thereof, in accordance with some embodiments. The seed layer **132** is formed using a deposition process, such as a physical vapor deposition process or a chemical vapor deposition process, in accordance with some embodiments.

(38) As shown in FIG. **1B**, a mask layer **134** is formed over the seed layer **132** over the dielectric layer **120**, in accordance with some embodiments. The mask layer **134** has openings **134a**, **134b** and **134c**, in accordance with some embodiments. The openings **134a**, **134b** and **134c** are respectively over portions of the seed layer **132** over the pads **115a** and **115b** and the dummy pads **115c**, in accordance with some embodiments. The mask layer **134** is made of a polymer material, such as a photoresist material, in accordance with some embodiments.

(39) As shown in FIG. **1B**, a conductive layer **136** is formed in the openings **122**, **124**, and **126** and over the seed layer **132**, in accordance with some embodiments. The conductive layer **136** is made of a conductive material, such as metal (e.g., copper, aluminum, gold, silver, or tungsten) or alloys thereof, in accordance with some embodiments. The conductive layer **136** is formed using a plating process, such as an electroplating process, in accordance with some embodiments.

(40) As shown in FIG. **1C**, the mask layer **134**, the seed layer **132** and the conductive layer **136** outside of the openings **122**, **124**, and **126** are removed, in accordance with some embodiments. The seed layer **132** and the conductive layer **136** remaining in the openings **122** form conductive vias **130a**, in accordance with some embodiments.

(41) The seed layer **132** and the conductive layer **136** remaining in the openings **124** form conductive vias **130b**, in accordance with some embodiments. The seed layer **132** and the conductive layer **136** remaining in the openings **126** form dummy vias **130c**, in accordance with some embodiments.

(42) The conductive vias **130a** are over and connected to the pads **115a**, in accordance with some embodiments. The conductive vias **130b** are over and connected to the pads **115b**, in accordance with some embodiments. The dummy vias **130c** are over and connected to the dummy pads **115c**, in accordance with some embodiments. The conductive vias **130a** and **130b** and the dummy vias **130c** pass through the dielectric layer **120**, in accordance with some embodiments.

(43) FIG. **1C-1** is a top view of a region A of the redistribution structure of FIG. **1C**, in accordance with some embodiments. For the sake of simplicity, FIG. **1C-1** does not show the seed layer **132**. The region A of FIG. **1C** shows a cross-sectional view illustrating the redistribution structure **110** along a sectional line A-A in FIG. **1C-1**, in accordance with some embodiments.

(44) As shown in FIGS. **1C** and **1C-1**, the dummy vias **130c** surround the corresponding conductive via **130a**, in accordance with some embodiments. The width **W130c** of the dummy via **130c** is substantially equal to or greater than the width **W130a** of the conductive via **130a**, in accordance with some embodiments. In some other embodiments, the width **W130c** is less than the width **W130a**.

(45) In some embodiments, one of the dummy vias **130c** is between the conductive via **130a** and **130b**, and a distance **D3** between the conductive via **130a** and the dummy via **130c** is less than a distance **D4** between the conductive via **130b** and the dummy via **130c**. In some other embodiments, the distance **D3** is substantially equal to or greater than the distance **D4**. The distance **D3** ranges from about 55 μm to about 105 μm , in accordance with some embodiments. The distance **D4** ranges from about 55 μm to about 105 μm , in accordance with some embodiments.

(46) Each conductive via **130a** is electrically connected to the conductive line **114a** thereunder through the pad **115a** and the conductive via **116** therebetween, in accordance with some embodiments. In some embodiments, portions of the conductive lines **114a** are under the dummy vias **130c**. That is, the dummy vias **130c** partially overlap the conductive lines **114a** thereunder, in accordance with some embodiments.

(47) The dummy pad **115c** is between the dummy via **130c** and the conductive line **114a**, in accordance with some embodiments. The dummy vias **130c** are electrically insulated from the redistribution layers **112**, **113**, and **114** (including the conductive lines **114a**), in accordance with some embodiments.

(48) As shown in FIG. 1D, a seed layer **142** is conformally formed over the dielectric layer **120**, the conductive vias **130a** and **130b**, and the dummy vias **130c**, in accordance with some embodiments. The seed layer **142** is made of a conductive material, such as metal (e.g., copper, aluminum, gold, silver, or tungsten) or alloys thereof, in accordance with some embodiments. The seed layer **142** is formed using a deposition process, such as a physical vapor deposition process or a chemical vapor deposition process, in accordance with some embodiments.

(49) As shown in FIG. 1D, a mask layer **144** is formed over the seed layer **142** over the dielectric layer **120**, in accordance with some embodiments. The mask layer **144** has openings **144a**, **144b** and **144c**, in accordance with some embodiments. The openings **144a**, **144b** and **144c** are respectively over portions of the seed layer **142** over the conductive vias **130a** and **130b** and the dummy vias **130c**, in accordance with some embodiments.

(50) The width **W1** of the opening **144a** is substantially equal to the width **W2** of the opening **144b**, in accordance with some embodiments. The width **W3** of the opening **144c** is substantially equal to or greater than the width **W1** or **W2**, in accordance with some embodiments. In some other embodiments, the width **W3** is less than the width **W1** or **W2**. The mask layer **144** is made of a polymer material, such as a photoresist material, in accordance with some embodiments.

(51) As shown in FIG. 1D, a conductive layer **148** is formed in the openings **144a**, **144b**, and **144c** and over the seed layer **142**, in accordance with some embodiments. The conductive layer **148** is made of a conductive material, such as metal (e.g., titanium, copper, nickel, or aluminum) or alloys thereof, in accordance with some embodiments. The conductive layer **148** is formed using a plating process, such as an electroplating process, in accordance with some embodiments.

(52) When the widths **W1** and **W2** of the openings **144a** and **144b** both are less than the width **W3** of the opening **144c**, the plating current density applied on the seed layer **142** under the openings **144a** and **144b** is greater than the plating current density applied on the seed layer **142** under the opening **144c**, in accordance with some embodiments. Therefore, the thickness **T1** of the conductive layer **148** in the opening **144a** and the thickness **T2** of the conductive layer **148** in the opening **144b** both are greater than the thickness **T3** of the conductive layer **148** in the opening **144c**, in accordance with some embodiments.

(53) As shown in FIG. 1E, the mask layer **144** and the seed layer **142** thereunder are removed, in accordance with some embodiments. The seed layer **142** and the conductive layer **148** remaining over the conductive vias **130a** together form conductive pillars **140a**, in accordance with some embodiments. The seed layer **142** and the conductive layer **148** remaining over the conductive vias **130b** together form conductive pillars **140b**, in accordance with some embodiments.

(54) The seed layer **142** and the conductive layer **148** remaining over the conductive vias **130c** together form dummy conductive pillars **140c**, in accordance with some embodiments. In some embodiments, each of the conductive pillar **140a** or **140b** or the dummy conductive pillar **140c** includes an under bump metallization layer.

(55) In some embodiments, each of the conductive pillar **140a** or **140b** or the dummy conductive pillar **140c** includes an under bump metallization layer and a conductive bump over the under bump metallization layer. In some embodiments, the width **W4** of the dummy conductive pillar **140c** is substantially equal to or greater than the width **W6** of the conductive pillar **140a**. In some

other embodiments, the width W4 is less than the width W6.

(56) In some embodiments, a dummy conductive pillar **140c** is between the conductive pillars **140a** and **140b**, and a distance D5 between the conductive pillar **140a** and the dummy conductive pillar **140c** is less than a distance D6 between the dummy conductive pillar **140c** and the conductive pillar **140b**. In some other embodiments, the distance D5 is substantially equal to or greater than the distance D6. The distance D5 ranges from about 25 μm to about 75 μm , in accordance with some embodiments. The distance D6 ranges from about 25 μm to about 75 μm , in accordance with some embodiments.

(57) As shown in FIGS. **1D** and **1E**, since the thickness T1 and the thickness T2 both are greater than the thickness T3, the thickness T1' of the conductive pillar **140a** and the thickness T2' of the conductive pillar **140b** both are greater than the thickness T3' of the dummy conductive pillar **140c**, in accordance with some embodiments. That is, top surfaces of the conductive pillars **140a** and **140b** are higher than a top surface of the dummy conductive pillar **140c**, which helps the conductive pillars **140a** and **140b** bond with solder bumps in a subsequent bonding process and prevents the solder bumps from short-circuiting to adjacent dummy conductive pillars **140c**, in accordance with some embodiments.

(58) FIG. **1E-1** is a top view of a right region A of the redistribution structure **110** of FIG. **1E**, in accordance with some embodiments. The region A of FIG. **1E** shows a cross-sectional view illustrating the redistribution structure **110** along a sectional line A-A in FIG. **1E-1**, in accordance with some embodiments.

(59) As shown in FIGS. **1E** and **1E-1**, the width W6 of the conductive pillar **140a** is less than the width W115a of the pad **115a**, which reserves more space for layout, in accordance with some embodiments. In some other embodiments, the width W6 is greater than or equal to the width W115a, which improves the ability of the conductive pillar **140a** to maintain balance in subsequent thermal processes. As shown in FIGS. **1E** and **1E-1**, the width W4 of the dummy conductive pillar **140c** is substantially equal to the width W115c of the dummy pad **115c**. In some other embodiments, the width W4 is greater than or less than the width W115c.

(60) FIG. **1E-2** is a top view of a left region B of the redistribution structure **110** of FIG. **1E**, in accordance with some embodiments. The region B of FIG. **1E** shows a cross-sectional view illustrating the redistribution structure **110** along a sectional line B-B in FIG. **1E-2**, in accordance with some embodiments. In some other embodiments, as shown in FIGS. **1E** and **1E-2**, the width W4 of the dummy conductive pillar **140c** is greater than the width W115c' of the dummy pad **115c**.

(61) As shown in FIG. **1E**, the dummy conductive pillar **140c**, the dummy via **130c** thereunder, and the dummy pad **115c** thereunder together form an anchor structure An, in accordance with some embodiments. As shown in FIGS. **1E**, **1E-1** and **1E-2**, the anchor structures An surround the corresponding conductive pillar **140a**, the corresponding conductive via **130a**, and the corresponding pad **115a**, in accordance with some embodiments. The anchor structures An together form an anchor ring-like structure R, in accordance with some embodiments.

(62) The distances D5, D7, D8, and D9 between different dummy conductive pillars **140c** and the corresponding conductive pillar **140a** are substantially the same, in accordance with some embodiments. In some other embodiments, at least two of the distances D5, D7, D8, and D9 are different from each other.

(63) As shown in FIG. **1F**, chip structures **150** are bonded to the conductive pillars **140a** through solder bumps **160**, in accordance with some embodiments. Each chip structure **150** includes a chip **152** and conductive pillars **154** over the chip **152**, in accordance with some embodiments. The chip structures **150** are electrically connected to the conductive vias **130a** thereunder through the solder bumps **160** and the conductive pillars **140a**, in accordance with some embodiments.

(64) The chip structures **150** are electrically insulated from the dummy vias **130c** and the dummy conductive pillars **140c**, in accordance with some embodiments. In some embodiments, the dummy conductive pillar **140c** between the conductive pillars **140a** and **140b** is also between the chip

structure **150** and the dummy via **130c** thereunder.

(65) In some embodiments, the chip **152** includes a substrate, devices, and an interconnect structure over the substrate, in accordance with some embodiments. The substrate, the devices, and the interconnect layers are not shown in figures for the purpose of simplicity and clarity.

(66) The substrate is made of an elementary semiconductor material including silicon or germanium in a single crystal structure, a polycrystal structure, or an amorphous structure. In some other embodiments, the substrate is made of a compound semiconductor, such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, an alloy semiconductor, such as SiGe, or GaAsP, or a combination thereof. The substrate may also include multi-layer semiconductors, semiconductor on insulator (SOI) (such as silicon on insulator or germanium on insulator), or a combination thereof.

(67) In some embodiments, the devices of the chip **152** are formed in and/or over the substrate. Examples of the various devices include active devices, passive devices, other suitable elements, or a combination thereof. The active devices may include transistors or diodes formed at a surface of the substrate **110**. The passive devices include resistors, capacitors, or other suitable passive devices.

(68) For example, the transistors may be metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high-voltage transistors, high-frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), etc. Various processes, such as front-end-of-line (FEOL) semiconductor fabrication processes, are performed to form the various device elements. The FEOL semiconductor fabrication processes may include deposition, etching, implantation, photolithography, annealing, planarization, one or more other applicable processes, or a combination thereof.

(69) In some embodiments, isolation features (not shown) are formed in the substrate. The isolation features are used to define active regions and electrically isolate various devices formed in and/or over the substrate in the active regions. In some embodiments, the isolation features include shallow trench isolation (STI) features, local oxidation of silicon (LOCOS) features, other suitable isolation features, or a combination thereof. The conductive pillars **154** are made of a conductive material, such as metal (e.g., titanium, copper, nickel, aluminum, gold, silver, or tungsten) or alloys thereof, in accordance with some embodiments.

(70) The interconnect structure is formed over the devices and the substrate, in accordance with some embodiments. The interconnect structure includes a dielectric layer, wiring layers, and conductive vias, in accordance with some embodiments. The wiring layers and the conductive vias are in the dielectric layer, in accordance with some embodiments.

(71) The conductive vias are electrically connected between the wiring layers, the devices, and the conductive pillars **154**, in accordance with some embodiments. Therefore, the conductive pillars **154** are electrically connected to the devices through the wiring layers and the conductive vias, in accordance with some embodiments.

(72) The dielectric layer is made of an oxide-containing material (e.g., silicon oxide or tetraethyl orthosilicate (TEOS) oxide), an oxynitride-containing material (e.g., silicon oxynitride), a glass material (e.g., borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), or fluorinated silicate glass (FSG)), or a combination thereof, in accordance with some embodiments.

(73) Alternatively, the dielectric layer includes a low-k material or a porous dielectric material having a k-value which is lower than that of silicon oxide, or lower than about 3.0 or about 2.5, in accordance with some embodiments. The wiring layers and the conductive vias are made of a conductive material, such as metal (e.g., copper, aluminum, gold, silver, or tungsten) or alloys thereof, in accordance with some embodiments. The solder bumps **160** are made of a conductive material, such as a tin-based alloy, in accordance with some embodiments.

(74) As shown in FIG. 1F, an underfill layer **170** is formed between the chip structures **150** and the redistribution structure **110**, in accordance with some embodiments. The underfill layer **170** surrounds the chip structures **150**, the solder bumps **160**, the conductive pillars **140a** and **140b**, and the dummy conductive pillars **140c**, in accordance with some embodiments.

(75) The underfill layer **170** extends into a gap **G1** between the chips **152** of the chip structures **150**, in accordance with some embodiments. The gap **G1** is filled with the underfill layer **170**, in accordance with some embodiments. The underfill layer **170** is made of an insulating material, such as a polymer material, in accordance with some embodiments.

(76) As shown in FIG. 1F, the molding layer **180** is formed over the redistribution structure **110** and the underfill layer **170**, in accordance with some embodiments. The molding layer **180** surrounds the chip structures **150** and the underfill layer **170**, in accordance with some embodiments. The molding layer **180** is made of an insulating material, such as a polymer material (e.g., epoxy), in accordance with some embodiments.

(77) As shown in FIG. 1G, portions of the dielectric layer **111** under the redistribution layer **112** are removed to form openings **111a** in the dielectric layer **111**, in accordance with some embodiments. The openings **111a** expose portions of the redistribution layer **112**, in accordance with some embodiments. The removal process includes an etching process, such as an anisotropic etching process (e.g., a dry etching process), in accordance with some embodiments.

(78) As shown in FIG. 1G, conductive pillars **190** are formed over the portions of the redistribution layer **112** exposed by the openings **111a**, in accordance with some embodiments. The conductive pillars **190** are electrically connected to the redistribution layer **112**, in accordance with some embodiments.

(79) The conductive pillars **190** are made of a conductive material, such as metal (e.g., copper, aluminum, gold, silver, tungsten, cobalt, nickel, or tin) or alloys thereof, in accordance with some embodiments. The conductive pillars **190** are formed using a plating process, such as an electroplating process, in accordance with some embodiments.

(80) As shown in FIG. 1G, solder bumps **210** are formed over the conductive pillars **190**, in accordance with some embodiments. The solder bumps **210** are made of a conductive material, such as a tin-based alloy, in accordance with some embodiments. The solder bumps **210** are made of a conductive material, such as a tin-based alloy, or another suitable conductive material with a melting point lower than that of the conductive pillars **190**, in accordance with some embodiments. The solder bumps **210** are formed using a plating process, such as an electroplating process, and a reflow process, in accordance with some embodiments.

(81) In this step, an interposer substrate **10** is substantially formed, in accordance with some embodiments. The interposer substrate **10** includes the redistribution structure **110**, the dielectric layer **120**, the conductive vias **130a** and **130b**, and the dummy vias **130c**, in accordance with some embodiments.

(82) As shown in FIG. 1G, a cutting process is performed to cut the interposer substrate **10** and the molding layer **180** along the cutting lines **C** to form chip package structures **100**, in accordance with some embodiments. The process of FIGS. 1A-1G is a wafer level packaging process, in accordance with some embodiments. For the sake of simplicity, FIGS. 1A-1G only show the structure for forming one of the chip package structures **100**, in accordance with some embodiments.

(83) As shown in FIG. 1G, top surfaces **131a** and **131b** of the conductive vias **130a** and **130b** and top surfaces **131c** of the dummy vias **130c** are substantially coplanar, in accordance with some embodiments. The top surfaces **131a**, **131b** and **131c** and a top surface **128** of the dielectric layer **120** are substantially coplanar, in accordance with some embodiments.

(84) As shown in FIG. 1H, a wiring substrate **220** is provided, in accordance with some embodiments. The wiring substrate **220** includes a dielectric layer **222**, bonding pads **224**, wiring layers **226**, and conductive vias **228**, in accordance with some embodiments. The bonding pads **224**

are formed over the dielectric layer **222**, in accordance with some embodiments. The wiring layers **226** and the conductive vias **228** are formed in the dielectric layer **222**, in accordance with some embodiments.

(85) The conductive vias **228** are electrically connected between different wiring layers **226** and between the wiring layer **226** and the bonding pads **224**, in accordance with some embodiments. For the sake of simplicity, FIG. **1H** only shows two of the wiring layers **226**, in accordance with some embodiments.

(86) The dielectric layer **222** is made of an insulating material such as a polymer material (e.g., polybenzoxazole, polyimide, or a photosensitive material), nitride (e.g., silicon nitride), oxide (e.g., silicon oxide), silicon oxynitride, or the like, in accordance with some embodiments. The dielectric layer **222** is formed using deposition processes (e.g. chemical vapor deposition processes or physical vapor deposition processes), photolithography processes, and etching processes, in accordance with some embodiments.

(87) The bonding pads **224** are made of a conductive material, such as metal (e.g. copper, aluminum, or tungsten) or alloys thereof, in accordance with some embodiments. The wiring layers **226** are made of a conductive material, such as metal (e.g. copper, aluminum, or tungsten) or alloys thereof, in accordance with some embodiments. The conductive vias **228** are made of a conductive material, such as metal (e.g. copper, aluminum, or tungsten) or alloys thereof, in accordance with some embodiments.

(88) In some embodiments, the bonding pads **224**, the wiring layers **226**, and the conductive vias **228** are made of the same material. In some other embodiments, the bonding pads **224**, the wiring layers **226**, and the conductive vias **228** are made of different materials. In some embodiments, the wiring substrate **220** includes a multilayer core substrate.

(89) As shown in FIG. **1H**, the chip package structure **100** is bonded to the wiring substrate **220** through the solder bumps **210**, in accordance with some embodiments. The solder bumps **210** are connected between the conductive pillars **190** and the bonding pads **224**, in accordance with some embodiments.

(90) The chip structures **150** are electrically connected to the wiring substrate **220** through the solder bumps **160**, the conductive pillars **140a** and **140b**, the conductive vias **130a**, **130b**, and **116**, the redistribution layers **112**, **113**, and **114**, the pads **115a** and **115b**, and the conductive vias (not shown) between the redistribution layers **112**, **113**, and **114**, in accordance with some embodiments.

(91) The anchor structures **An** (each including the dummy conductive pillar **140c**, the dummy via **130c** thereunder, and the dummy pad **115c** thereunder) are electrically insulated from the chip structures **150**, the wiring substrate **220**, the solder bumps **160**, the conductive pillars **140a** and **140b**, the conductive vias **130a**, **130b**, and **116**, the redistribution layers **112**, **113**, and **114**, the pads **115a** and **115b**, and the conductive vias (not shown) between the redistribution layers **112**, **113**, and **114**, in accordance with some embodiments.

(92) As shown in FIG. **1H**, an underfill layer **230** is formed between the chip package structure **100** and the wiring substrate **220**, in accordance with some embodiments. The underfill layer **230** surrounds the solder bumps **210** and the conductive pillars **190**, in accordance with some embodiments. The underfill layer **230** is made of an insulating material, such as a polymer material, in accordance with some embodiments.

(93) FIG. **1H-1** is a top view of the chip package structure of FIG. **1H**, in accordance with some embodiments. FIG. **1H** is a cross-sectional view illustrating the chip package structure along a sectional line I-I' in FIG. **1H-1**, in accordance with some embodiments.

(94) As shown in FIGS. **1H** and **1H-1**, an adhesive layer **240** is formed over the wiring substrate **220**, in accordance with some embodiments. The adhesive layer **240** has an opening **242**, in accordance with some embodiments. The chip package structure **100** is in the opening **242**, in accordance with some embodiments. The adhesive layer **240** has a ring shape, in accordance with some embodiments. The adhesive layer **240** is made of a polymer material such as epoxy or

silicone, in accordance with some embodiments.

(95) As shown in FIGS. 1H and 1H-1, a ring structure **250** is disposed over the adhesive layer **240**, in accordance with some embodiments. In this step, a chip package structure **200** is substantially formed, in accordance with some embodiments. The ring structure **250** has an opening **252** over the opening **242**, in accordance with some embodiments.

(96) The chip package structure **100** is in the opening **252**, in accordance with some embodiments. In some embodiments, the ring structure **250** and the adhesive layer **240** have substantially the same shape and substantially the same size. The ring structure **250** is made of a rigid material, such as metal (e.g., copper or iron), alloys thereof (e.g., stainless steel), or another suitable material which is more rigid than the wiring substrate **220**, in accordance with some embodiments.

(97) FIG. 1H-2 is a top view of the chip package structure of FIG. 1H, in accordance with some embodiments. For the sake of simplicity and clarity, FIG. 1H-2 does not show the ring structure **250**, the chip structures **150**, the underfill layers **170** and **230**, and the molding layer **180**, in accordance with some embodiments.

(98) As shown in FIGS. 1H-1 and 1H-2, the conductive pillars **140a**, the conductive vias **130a**, and the anchor structures An (or the dummy vias **130c**) are close to corners **10c** of the interposer substrate **10**, in accordance with some embodiments. As shown in FIG. 1H-2, the conductive pillars **140a**, the conductive vias **130a**, and the anchor structures An are in corner regions **10r** of the interposer substrate **10**, in accordance with some embodiments.

(99) As shown in FIG. 1H-2, the corner regions **10r** have a square shape, in accordance with some embodiments. In some other embodiments, as shown in FIG. 1H-3, the corner regions **10r** have a fan shape or a substantially quarter-round shape. The area of each corner region **10r** ranges from about 700 μm^2 to about 1300 μm^2 , in accordance with some embodiments.

(100) The conductive pillars **140a** and the conductive vias **130a** are also referred to as corner conductive pillars and corner conductive vias, in accordance with some embodiments. In some embodiments, as shown in FIG. 1H-2, the length L of the interposer substrate **10** is greater than about 26 mm, and the interposer substrate **10** is also referred to as a super large interposer substrate, in accordance with some embodiments.

(101) The thermal expansion coefficient of the chip structures **150** is different from (e.g., less than) that of the wiring substrate **220**, in accordance with some embodiments. The mismatch of thermal expansion coefficients between the chip structures **150** and the wiring substrate **220** may induce a thermal stress in the interposer substrate **10**, especially in the corner regions **10r** of the interposer substrate **10**, in subsequent thermal processes, in accordance with some embodiments. The thermal stress tends to result in cracks in the conductive vias **130a**, the pads **115a**, the conductive vias **116**, and the conductive lines **114a**, in accordance with some embodiments.

(102) The anchor structures An are made of metal or alloys thereof, the dielectric layers **111** and **120** are made of a polymer material or oxide, and therefore the anchor structures An are more rigid than the dielectric layers **111** and **120**, in accordance with some embodiments.

(103) Since the anchor structures An are more rigid than the dielectric layers **111** and **120**, the anchor structures An may constrain portions of the interposer substrate **10**, which is close to the anchor structures An, to reduce the stress and the strain in the interposer substrate **10** in the corner regions **10r** in subsequent thermal processes. Therefore, in subsequent thermal processes, the thermal stress in the conductive vias **130a**, the pads **115a**, the conductive vias **116**, and the conductive lines **114a** in the corner regions **10r** is reduced, in accordance with some embodiments. As a result, the anchor structures An improve the reliability of the chip package structure **200**, in accordance with some embodiments. The formation process of the anchor structures An is compatible with the formation process of the conductive vias **130a** and the pads **115a**, and therefore no additional manufacturing cost is needed and no additional process time is required, in accordance with some embodiments.

(104) In some embodiments, the conductive pillar **140a**, the conductive via **130a** thereunder, and

the pad **115a** thereunder together form a corner connector **C1**. In some embodiments, the conductive pillar **140b**, the conductive via **130b** thereunder, and the pad **115b** thereunder together form a connector **C2**.

(105) In some embodiments, the anchor structure **An** (between the corner connector **C1** and the connector **C2**) is closer to the corner connector **C1** and farther from the connector **C2**, which reduces the dielectric layer **120** between the anchor structure **An** and the corner connector **C1**. Therefore, the anchor structure **An** may constrain the dielectric layer **120** between the anchor structure **An** and the corner connector **C1** more efficiently to reduce more stress and strain in the dielectric layer **120** therebetween in subsequent thermal processes.

(106) When the anchor structure **An** is closer to the corner connector **C1** and farther from the connector **C2**, the distance **D1** between the pad **115a** and the dummy pad **115c** is less than the distance **D2** between the dummy pad **115c** and the pad **115b**, the distance **D3** between the conductive via **130a** and the dummy via **130c** is less than the distance **D4** between the conductive via **130b** and the dummy via **130c**, and the distance **D5** between the conductive pillar **140a** and the dummy conductive pillar **140c** is less than the distance **D6** between the dummy conductive pillar **140c** and the conductive pillar **140b**, in accordance with some embodiments.

(107) In some other embodiments, the anchor structure **An** is closer to the connector **C2** and farther from the corner connector **C1**, which reduces the dielectric layer **120** between the anchor structure **An** and the connector **C2**. Therefore, the anchor structure **An** may constrain the dielectric layer **120** between the anchor structure **An** and the connector **C2** more efficiently to reduce more stress and strain in the dielectric layer **120** therebetween in subsequent thermal processes. As a result, the anchor structure **An** may protect the connector **C2** from damage in subsequent thermal processes.

(108) When the anchor structure **An** is closer to the connector **C2** and farther from the corner connector **C1**, the distance **D1** between the pad **115a** and the dummy pad **115c** is greater than the distance **D2** between the dummy pad **115c** and the pad **115b**, the distance **D3** between the conductive via **130a** and the dummy via **130c** is greater than the distance **D4** between the conductive via **130b** and the dummy via **130c**, and the distance **D5** between the conductive pillar **140a** and the dummy conductive pillar **140c** is greater than the distance **D6** between the dummy conductive pillar **140c** and the conductive pillar **140b**, in accordance with some embodiments.

(109) In still other embodiments, the anchor structure **An** is positioned at the middle between the connector **C2** and the corner connector **C1**, and therefore the anchor structure **An** equally constrains the dielectric layer **120** between the anchor structure **An** and the connector **C2** and between the anchor structure **An** and the corner connector **C1** to reduce stress and strain in the dielectric layer **120** therebetween in subsequent thermal processes. As a result, the anchor structure **An** may protect both the corner connector **C1** and the connector **C2** from damage in subsequent thermal processes.

(110) When the anchor structure **An** is positioned at the middle between the connector **C2** and the corner connector **C1**, the distance **D1** between the pad **115a** and the dummy pad **115c** is equal to the distance **D2** between the dummy pad **115c** and the pad **115b**, the distance **D3** between the conductive via **130a** and the dummy via **130c** is equal to the distance **D4** between the conductive via **130b** and the dummy via **130c**, and the distance **D5** between the conductive pillar **140a** and the dummy conductive pillar **140c** is equal to the distance **D6** between the dummy conductive pillar **140c** and the conductive pillar **140b**, in accordance with some embodiments.

(111) In some embodiments, the anchor structure **An** is wider than or equal to the corner connector **C1**, which improves the ability of the anchor structure **An** to maintain balance in subsequent thermal processes. Therefore, the anchor structure **An** may constrain the dielectric layer **120** between the anchor structure **An** and the corner connector **C1** more efficiently to reduce more stress and strain in the dielectric layer **120** therebetween in subsequent thermal processes.

(112) When the anchor structure **An** is wider than or equal to the corner connector **C1**, the width **W115c** of the dummy pad **115c** is greater than or substantially equal to the width **W115a** of the pad **115a**, the width **W130c** of the dummy via **130c** is greater than or substantially equal to the width

W130a of the conductive via 130a, and the width W4 of the dummy conductive pillar 140c is greater than or substantially equal to the width W6 of the conductive pillar 140a, in accordance with some embodiments.

(113) In some embodiments, the anchor structure An is narrower than the corner connector C1, which reserves more space for circuit layout. When the anchor structure An is narrower than the corner connector C1, the width W115c of the dummy pad 115c is less than the width W115a of the pad 115a, the width W130c of the dummy via 130c is less than the width W130a of the conductive via 130a, and the width W4 of the dummy conductive pillar 140c is less than the width W6 of the conductive pillar 140a, in accordance with some embodiments.

(114) FIGS. 1H, 1H-1 and 1H-2 show the anchor structures An are in the corner regions 10r, though the present invention is not limited thereto. For example, the anchor structures An may surround the conductive vias 130b and the conductive pillars 140b according to requirements.

(115) Although FIG. 1H-2 shows four anchor structures An surround one conductive via 130a, the present invention is not limited thereto. The number of the anchor structures An surrounding one conductive via 130a may be varied. In some embodiments, as shown in FIG. 2, the number of the anchor structures An surrounding one conductive via 130a is two. The two anchor structures An are on opposite sides of the conductive via 130a, in accordance with some embodiments.

(116) In some embodiments, as shown in FIG. 3, the number of the anchor structures An surrounding one conductive via 130a is three. The three anchor structures An are arranged at vertices of a triangle, in accordance with some embodiments. The conductive via 130a is arranged at a center of the triangle, in accordance with some embodiments.

(117) In some embodiments, as shown in FIG. 4, the number of the anchor structures An surrounding one conductive via 130a is five. The five anchor structures An are arranged at vertices of a pentagon, in accordance with some embodiments. The conductive via 130a is arranged at a center of the pentagon, in accordance with some embodiments.

(118) In some embodiments, as shown in FIG. 5, the number of the anchor structures An surrounding one conductive via 130a is six. The six anchor structures An are arranged at vertices of a hexagon, in accordance with some embodiments. The conductive via 130a is arranged at a center of the hexagon, in accordance with some embodiments.

(119) In some embodiments, as shown in FIG. 6, only one anchor structure An continuously surrounds one conductive via 130a. The anchor structure An forms the anchor ring-like structure R, in accordance with some embodiments. The anchor structure An has a ring shape, in accordance with some embodiments. That is, the dummy via 130c and the dummy conductive pillar 140c have a ring shape, in accordance with some embodiments.

(120) The anchor structure An continuously surrounds the entire conductive via 130a and the conductive pillar 140a, in accordance with some embodiments. In some embodiments, a width W130c of the dummy via 130c is substantially equal to or greater than a width W130a of the conductive via 130a. In some other embodiments, the width W130c is less than the width W130a.

(121) In some embodiments, as shown in FIG. 7, the number of the anchor structures An surrounding one conductive via 130a is two. The two anchor structures An are spaced apart from each other by gaps G3, in accordance with some embodiments. The two anchor structures An together form the anchor ring-like structure R, in accordance with some embodiments. Each anchor structure An has a recess An1, in accordance with some embodiments. The conductive via 130a is partially in the recess An1, in accordance with some embodiments.

(122) In some embodiments, as shown in FIG. 8, the number of the anchor structures An surrounding one conductive via 130a is three. The three anchor structures An are spaced apart from each other by gaps G4, in accordance with some embodiments. The three anchor structures An together form the anchor ring-like structure R, in accordance with some embodiments.

(123) FIG. 9 is a cross-sectional view illustrating a chip package structure 900, in accordance with some embodiments. As shown in FIG. 9, the chip package structure 900 is similar to the chip

package structure **200** of FIG. **1H**, except that the thickness **T1'** of the conductive pillar **140a** or the thickness **T2'** of the conductive pillar **140b** is substantially equal to the thickness **T3'** of the dummy conductive pillar **140c**, in accordance with some embodiments. That is, top surfaces of the conductive pillars **140a** and **140b** and the dummy conductive pillar **140c** are substantially level with each other, in accordance with some embodiments.

(124) In some other embodiments, the thickness **T1'** or **T2'** is less than the thickness **T3'**, which reduces the distance between the chip structures **150** and the interposer substrate **10** to reduce the thickness of the chip package structure **100** so as to minimize the size of the chip package structure **900**. The thicknesses **T1'**, **T2'**, and **T3'** are adjustable according to requirements, in accordance with some embodiments. The thicknesses **T1'**, **T2'**, and **T3'** may be adjusted by adjusting process parameters of the plating process for forming the conductive pillars **140a** and the dummy conductive pillar **140c**, in accordance with some embodiments.

(125) The thickness **T1** of the conductive layer **148** of the conductive pillar **140a** is equal to or less than the thickness **T3** of the conductive layer **148** of the dummy conductive pillar **140c**, in accordance with some embodiments. The thickness **T2** of the conductive layer **148** of the conductive pillar **140b** is equal to or less than the thickness **T3**, in accordance with some embodiments.

(126) FIGS. **10A-10B** are cross-sectional views of various stages of a process for forming a chip package structure, in accordance with some embodiments. As shown in FIG. **10A**, after the step of FIG. **1A**, a dielectric layer **1010** is formed over the redistribution structure **110**, in accordance with some embodiments.

(127) The dielectric layer **1010** has openings **1012**, **1014**, and **1016**, in accordance with some embodiments. The openings **1012** expose the pads **115a** thereunder, in accordance with some embodiments. The openings **1014** expose the pads **115b** thereunder, in accordance with some embodiments. The openings **1016** expose the pads **115c** thereunder, in accordance with some embodiments.

(128) The dielectric layer **1010** is made of an insulating material such as a polymer material (e.g., polybenzoxazole, polyimide, or a photosensitive material), nitride (e.g., silicon nitride), oxide (e.g., silicon oxide), silicon oxynitride, or the like, in accordance with some embodiments. The dielectric layer **1010** is formed using deposition processes (e.g. chemical vapor deposition processes or physical vapor deposition processes), photolithography processes, and etching processes, in accordance with some embodiments.

(129) FIG. **10A-1** is a top view of a corner region of the interposer substrate of FIG. **10A**, in accordance with some embodiments. As shown in FIGS. **10A** and **10A-1**, conductive vias **1022a** and **1022b** and dummy vias **1022c** are respectively formed in the openings **1012**, **1014** and **1016**, and a redistribution layer **1024** is formed over the dielectric layer **1010** and the conductive vias **1022a** and **1022b** and the dummy vias **1022c**, in accordance with some embodiments.

(130) The redistribution layer **1024** includes pads **1024a** and **1024b** and dummy pads **1024c**, in accordance with some embodiments. The pads **1024a** and **1024b** and dummy pads **1024c** are respectively formed over the conductive vias **1022a** and **1022b** and the dummy vias **1022c**, in accordance with some embodiments.

(131) The conductive vias **1022a** and **1022b** and dummy vias **1022c** and the redistribution layer **1024** are made of a conductive material, such as metal (e.g. copper, aluminum, or tungsten) or alloys thereof, in accordance with some embodiments. The conductive vias **1022a** and **1022b** and dummy vias **1022c** and the redistribution layer **1024** are formed using a plating process, such as an electroplating process, in accordance with some embodiments.

(132) As shown in FIG. **10B**, the steps of FIGS. **1B-1H** are performed to form the dielectric layer **120**, the conductive vias **130a** and **130b**, the dummy vias **130c**, the conductive pillars **140a** and **140b**, the dummy conductive pillars **140c**, the chip structures **150**, the solder bumps **160**, the underfill layer **170**, the molding layer **180**, the conductive pillars **190**, the solder bumps **210**, the

wiring substrate **220**, the underfill layer **230**, the adhesive layer **240**, and the ring structure **250** of the chip package structure **200** of FIG. **1H**, in accordance with some embodiments.

(133) The dielectric layer **120** is formed over the dielectric layer **1010** and the redistribution layer **1024**, in accordance with some embodiments. The conductive vias **130a** and **130b** and the dummy vias **130c** are in the dielectric layer **120** and respectively over the pads **1024a** and **1024b** and the dummy pads **1024c**, in accordance with some embodiments.

(134) The arrangements of the conductive pillars **140a** and **140b**, the dummy conductive pillars **140c**, the chip structures **150**, the solder bumps **160**, the underfill layer **170**, the molding layer **180**, the conductive pillars **190**, the solder bumps **210**, the wiring substrate **220**, the underfill layer **230**, the adhesive layer **240**, and the ring structure **250** are substantially the same or similar to that of the chip package structure **200** of FIG. **1H**, in accordance with some embodiments. In this step, a chip package structure **1000** is substantially formed, in accordance with some embodiments.

(135) Processes and materials for forming the chip package structures **900** and **1000** may be similar to, or the same as, those for forming the chip package structure **200** described above.

(136) In accordance with some embodiments, chip package structures and methods for forming the same are provided. The methods (for forming the chip package structure) form anchor structures in a corner region of an interposer substrate which is between chip structures and a wiring substrate. The anchor structures are able to constrain the interposer substrate in the corner region so as to prevent conductive vias and conductive lines in the corner region from damage caused by a thermal stress in the interposer substrate, which is induced by the mismatch of thermal expansion coefficients between the chip structures and the wiring substrate. Therefore, the anchor structures improve the reliability of the chip package structure.

(137) In accordance with some embodiments, a chip package structure is provided. The chip package structure includes a wiring substrate. The chip package structure includes an interposer substrate over the wiring substrate. The interposer substrate includes a redistribution structure, a dielectric layer, a conductive via, and a plurality of first dummy vias, the dielectric layer is over the redistribution structure, the conductive via and the first dummy vias pass through the dielectric layer, the first dummy vias surround the conductive via, and the first dummy vias are electrically insulated from the wiring substrate. The chip package structure includes a chip structure over the interposer substrate. The chip structure is electrically connected to the conductive via, and the chip structure is electrically insulated from the first dummy vias.

(138) In accordance with some embodiments, a chip package structure is provided. The chip package structure includes an interposer substrate including a redistribution structure, a dielectric layer, and a conductive via. The dielectric layer is over the redistribution structure, and the conductive via is in the dielectric layer and electrically connected to the redistribution structure. The chip package structure includes an anchor ring-like structure partially embedded in the dielectric layer and surrounding the conductive via. The anchor ring-like structure is more rigid than the dielectric layer. The chip package structure includes a chip structure over the interposer substrate. The chip structure is electrically connected to the conductive via, and the chip structure is electrically insulated from the anchor ring-like structure.

(139) In accordance with some embodiments, a method for forming a chip package structure is provided. The method includes forming a dielectric layer over a redistribution structure. The method includes forming a first conductive via and a plurality of dummy vias in the dielectric layer. The dummy vias surround the first conductive via. The method includes bonding a chip structure to the redistribution structure. The chip structure is electrically connected to the first conductive via, and the chip structure is electrically insulated from the dummy vias. The method includes bonding the redistribution structure to a wiring substrate. The dummy vias are electrically insulated from the wiring substrate.

(140) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate

that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for forming a chip package structure, comprising: forming a dielectric layer over a redistribution structure; forming a first conductive via and a plurality of dummy vias in the dielectric layer, wherein the dummy vias surround the first conductive via; forming a conductive pillar over the first conductive via; forming a plurality of dummy conductive pillars over the dummy vias; bonding a chip structure to the redistribution structure, wherein the chip structure is electrically connected to the first conductive via, the chip structure is electrically insulated from the dummy vias, the chip structure is bonded to the conductive pillar, and the chip structure is electrically insulated from the dummy conductive pillars; and bonding the redistribution structure to a wiring substrate, wherein the dummy vias are electrically insulated from the wiring substrate, and the redistribution structure is between the dielectric layer and the wiring substrate and between the chip structure and the wiring substrate.
2. The method for forming the chip package structure as claimed in claim 1, wherein the forming of the first conductive via and the dummy vias in the dielectric layer further comprises: forming a second conductive via in the dielectric layer, wherein one of the dummy vias is between the first conductive via and the second conductive via.
3. The method for forming the chip package structure as claimed in claim 2, wherein a first distance between the first conductive via and the one of the dummy vias is less than a second distance between the second conductive via and the one of the dummy vias.
4. The method for forming the chip package structure as claimed in claim 1, wherein a first width of one of the dummy vias is greater than a second width of the first conductive via.
5. The method for forming the chip package structure as claimed in claim 4, wherein the first width of the one of the dummy vias is greater than a distance between the first conductive via and the one of the dummy vias.
6. The method for forming the chip package structure as claimed in claim 1, wherein a first top surface of the first conductive via and second top surfaces of the dummy vias are substantially coplanar.
7. The method for forming the chip package structure as claimed in claim 6, wherein the first top surface, the second top surfaces, and a third top surface of the dielectric layer are substantially coplanar.
8. The method for forming the chip package structure as claimed in claim 1, wherein one of the dummy conductive pillars is wider than the conductive pillar.
9. The method for forming the chip package structure as claimed in claim 1, wherein one of the dummy conductive pillars is between the chip structure and one of the dummy vias.
10. The method for forming the chip package structure as claimed in claim 1, wherein the redistribution structure comprises: a conductive line partially under the first conductive via and at least one of the dummy vias; and a second conductive via connected between the first conductive via and the conductive line.
11. The method for forming the chip package structure as claimed in claim 1, wherein the conductive pillar is thicker than one of the dummy conductive pillars.
12. The method for forming the chip package structure as claimed in claim 1, wherein the dummy conductive pillars are between the chip structure and the redistribution structure.

13. The method for forming the chip package structure as claimed in claim 12, wherein a first top surface of the dummy conductive pillar is lower than a second top surface of the conductive pillar.

14. A method for forming a chip package structure, comprising: forming a dielectric layer over a redistribution structure, wherein the redistribution structure comprises a redistribution layer, and the redistribution layer comprises a pad and a dummy pad, and the dummy pad is wider than the pad; forming a conductive via and a first dummy via in the dielectric layer, wherein the first dummy via surrounds the conductive via, the conductive via is over and connected to the pad, and the first dummy via is over and connected to the dummy pad; bonding a chip structure to the redistribution structure, wherein the chip structure is electrically connected to the conductive via, and the chip structure is electrically insulated from the first dummy via; and bonding the redistribution structure to a wiring substrate, wherein the first dummy via is electrically insulated from the wiring substrate.

15. The method for forming the chip package structure as claimed in claim 14, wherein the first dummy via has a ring shape.

16. The method for forming the chip package structure as claimed in claim 14, further comprising: forming a second dummy via in the dielectric layer, wherein the second dummy via surrounds the conductive via, the first dummy via and the second dummy via are spaced apart from each other, and the first dummy via and the second dummy via together form a ring structure.

17. The method for forming the chip package structure as claimed in claim 14, wherein a first width of the first dummy via is substantially equal to or greater than a second width of the conductive via.

18. A method for forming a chip package structure, comprising: forming a conductive pillar and dummy conductive pillars over a redistribution structure, wherein the dummy conductive pillars surround the conductive pillar, and the dummy conductive pillars are thinner and wider than the conductive pillar; bonding a chip structure to the redistribution structure, wherein the chip structure is electrically connected to the conductive pillar, and the chip structure is electrically insulated from the dummy conductive pillars; forming an underfill layer between the chip structure and the redistribution structure, wherein a portion of the underfill is between the chip structure and the dummy conductive pillars, and the portion of the underfill is in direct contact with top surfaces of the dummy conductive pillars; and bonding the redistribution structure to a wiring substrate, wherein the dummy conductive pillars are electrically insulated from the wiring substrate.

19. The method for forming the chip package structure as claimed in claim 18, wherein the conductive pillar and the dummy conductive pillars are under a corner portion of the chip structure.

20. The method for forming the chip package structure as claimed in claim 18, wherein the redistribution structure comprises a dielectric layer and a redistribution layer in the dielectric layer, and the dummy conductive pillars are more rigid than the dielectric layer.
